BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7483

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. When CO_2 is passed through the ammoniacal brine the only salt that precipitates is:

(A) $NaHCO_3$
(B) NH_4HCO_3
(C) Na_2CO_3
(D) $(NH_4)_2CO_3$

جب امونیکل برائن سے CO_2 گزارا جاتا ہے تو کونسا نمک رسوب بنتا ہے؟

Q2. Specific heat capacity of water is:

(A) $4.2kJg^{-1}K^{-1}$
(B) $4.2Jg^{-1}K^{-1}$
(C) $2.4kJg^{-1}K^{-1}$
(D) $2.4Jg^{-1}K^{-1}$

پانی کی مخصوص ہیٹ کیپیسٹی کتنی ہے؟

Q3. Temporary hardness is removed by adding:

(A) Quick lime
(B) Sodium bicarbonate
(C) Lime stone
(D) Lime water

ٹمپریری ہارڈنیس کو کونسا نمک ڈال کر ختم کیا جاتا ہے؟

(A) ان بجھا چونا
(B) سوڈیم بائی کاربونیٹ
(C) چونے کا پتھر
(D) چونے کا پانی

Q4. Height of mesosphere is about _____ kilometer from earth.

(A) 0-12
(B) 12-50
(C) 50-85
(D) 85-120

میسو سفیر کی زمین سے اونچائی _____ کلومیٹر ہے۔

Q5. Which gas protects the earth's surface from ultraviolet radiations?

(A) CO_2
(B) CO
(C) N_2
(D) O_3

کونسی گیس زمین کی سطح کو الٹرا وائلٹ ریڈی ایشنز سے محفوظ رکھتی ہے؟

Q6. Ozone is beneficial for us as it absorbs _____.

(A) Infrared radiations
(B) Ultraviolet radiations
(C) Chlorofluorocarbons
(D) Air pollutants

اوزون ہمارے لیے مفید ہے کیونکہ یہ _____ کو جذب کرتی ہے۔

(A) انفرا ریڈ ریڈی ایشنز
(B) الٹرا وائلٹ ریڈی ایشنز
(C) کلورو فلورو کاربنز
(D) ہوا کے پلوٹنٹس

Q7. Which is a fat soluble vitamin?

(A) Vitamin A
(B) Vitamin E
(C) Vitamin K
(D) All these

کونسا وٹامن فیٹ سولیبل ہے؟

(A) وٹامن A
(B) وٹامن E
(C) وٹامن K
(D) یہ تمام

Q8. The end product of oxidation of acetylene is:

(A) Oxalic acid
(B) Glycol
(C) Glyoxal
(D) None of these

ایسیٹیلین کی آکسیڈیشن کا آخری پراڈکٹ کونسا ہے؟

(A) آگزالک ایسڈ
(B) گلائی کول
(C) گلائی آکسل
(D) ان میں سے کوئی نہیں

Q9. The functional group -COOH is found in:

(A) Carboxylic acids
(B) Aldehydes
(C) Alcohols
(D) Esters

فنکشنل گروپ -COOH کس میں پایا جاتا ہے؟

(A) کاربوکسائلک ایسڈز
(B) ایلڈیہائیڈز
(C) الکوحلز
(D) ایسٹرز

Q10. Which is a synthetic fiber?

(A) Cotton
(B) Natural gas
(C) Nylon
(D) Petroleum

کونسا مصنوعی فائبر ہے؟

(A) کاٹن
(B) قدرتی گیس
(C) نائلون
(D) پٹرولیم

Q11. Which ion is the conjugate base of sulphuric acid?

(A) SO_3^{-2}
(B) S^{-2}
(C) HSO_3^{-}
(D) HSO_4^{-}

سلفیورک ایسڈ کا کنجوگیٹ بیس ہے:

Q12. For a reaction between PCl_3 and Cl_2 to form PCl_5, the units of Kc are:

(A) mol dm^{-3}
(B) mol^{-1} dm^{-3}
(C) mol^{-1} dm^3
(D) mol dm^3

PCl_3 اور Cl_2 سے PCl_5 بنانے کے لیے ری ایکشن میں Kc کے یونٹس ہیں:

Chemistry (Subjective) Group - I

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7483

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan]e between reversible and irreversible reactions

(ii) [illegible in source scan]on of a reaction can be predicted?

(iii) [illegible in source scan]tional group with an example.

(iv) [illegible in source scan]haracteristics of homologous series.

(v) [illegible in source scan] organic compounds are used as food.

(vi) [illegible in source scan] importance of natural gas?

(vii) [illegible in source scan] fatty acids with their formulae.

(viii) [illegible in source scan] synthesized carbohydrates? Write equation.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan]ic salt? Give an example. (1+1)

(ii) [illegible in source scan]enius base with an example. (1+1)

(iii) [illegible in source scan]mplete combustion of methane? Give its reaction. (1+1)

(iv) [illegible in source scan]es of chloroform. (1+1)

(v) [illegible in source scan]cal reaction in ammonia recovery tower.

(vi) [illegible in source scan]ulation of urea?

(vii) [illegible in source scan]osition and boiling range of fuel oil.

(viii) [illegible in source scan]nia is prepared by Haber process?

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan]ources of oxides of carbon. (1+1)

(ii) [illegible in source scan]onstituents of troposphere. (1+1)

(iii) [illegible in source scan]ffects of SO_2 gas. (1+1)

(iv) [illegible in source scan]mary or secondary air pollutants: (½×4) (a) SO_2 (b) CH_4 (c) HNO_3 (d) H_2SO_4

(v) [illegible in source scan] mean by boiler scales? How are they removed? (1+1)

(vi) [illegible in source scan] between soft water and hard water. (1+1)

(vii) [illegible in source scan]hysical properties of water. (1+1)

(viii) [illegible in source scan]ss of water is removed by Clark's method? Also write equation. (1+1)

Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) [illegible in source scan] vitamins? Describe water soluble vitamins. Write the importance of vitamins.

Q5(b). (Marks 05) [illegible in source scan]ilibrium constant. Explain its units with the given reaction. $N_2 + 3H_2 \rightleftharpoons 2NH_3$

Q6(a). (Marks 04) [illegible in source scan] are named? Describe characteristics properties of salts.

Q6(b). (Marks 05) [illegible in source scan]ter born diseases. Write measures for the prevention of water born diseases.

Q7(a). (Marks 04) [illegible in source scan] advantages of Solvay's process.

Q7(b). (Marks 05) [illegible in source scan] methods for the preparation of alkenes.

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7488

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Vitamins are accessory growth factors. They play important role in our body like:

(A) Provide energy to the body
(B) Insulate our body from electric shock
(C) Build brain cells
(D) Regulate metabolic process

وٹامنز گروتھ سے متعلق فیکٹرز ہیں۔ یہ جسم میں اہم کردار ادا کرتے ہیں جیسا کہ:

(A) جسم کو توانائی مہیا کرتے ہیں
(B) ہمارے جسم کو الیکٹرک شاک سے محفوظ رکھتے ہیں
(C) دماغ کے سیلز بناتے ہیں
(D) میٹابولک پراسس کو ریگولیٹ کرتے ہیں

Q2. The order of reactivity of hydrogen halides with alkenes is:

(A) HI > HBr
(B) HBr > HI
(C) HCl > HBr
(D) HBr < HCl

الکینز کے ساتھ ہائیڈروجن ہیلائیڈز کی ری ایکٹیویٹی کی ترتیب ہے:

Q3. Formula of acetaldehyde is:

(A) CH_3-CH_2OH
(B) CH_3-C(=O)-OH
(C) CH_3-C(=O)-H
(D) H-C(=O)-H

ایسیٹلڈیہائیڈ کا فارمولا کونسا ہے؟

Q4. Coal having 90% carbon content is called:

(A) Peat
(B) Lignite
(C) Anthracite
(D) Bituminous

جس کوئلے میں 90 فیصد کاربن کے اجزاء موجود ہوتے ہیں وہ کہلاتا ہے:

(A) پیٹ
(B) لگنائٹ
(C) اینتھرائٹ
(D) بٹیومینس

Q5. Which ion is the conjugate base of sulphuric acid?

(A) SO_3^{-2}
(B) S^{-2}
(C) HSO_4^{-1}
(D) HSO_3^{-1}

سلفیورک ایسڈ کا کنجوگیٹ بیس ہے:

Q6. In the lime kiln, the reaction goes to complete because $CaCO_3 \rightarrow CaO + CO_2$

(A) of high temperature
(B) CaO is more stable than $CaCO_3$
(C) CO_2 escapes continuously
(D) CaO is not dissociated

چونے کی بھٹی میں دیے گئے ری ایکشن کے مکمل ہونے کی وجہ ہے:

(A) زیادہ درجہ حرارت کی وجہ سے
(B) $CaCO_3$، CaO سے زیادہ مستحکم ہے
(C) CO_2 مسلسل خارج ہوتی رہتی ہے
(D) CaO ڈسوسی ایٹ نہیں ہوتا

Q7. Which is used as jet fuel?

(A) Kerosene oil
(B) Lubricating oil
(C) Fuel oil
(D) Diesel oil

کونسی فریکشن بطور جیٹ فیول استعمال ہوتی ہے؟

(A) کیروسین آئل
(B) لبریکیٹنگ آئل
(C) فیول آئل
(D) ڈیزل آئل

Q8. Which ion does not cause hardness in water?

(A) Ca^{+2}
(B) Mg^{+2}
(C) SO_4^{-2}
(D) Na^{+1}

کونسا آئن واٹر ہارڈنیس کی وجہ نہیں بنتا ہے؟

Q9. Rapid growth of algae in water bodies is because of detergents having:

(A) Carbonate salts
(B) Sulphonic acid salts
(C) Sulphate salts
(D) Phosphate salts

ڈیٹرجنٹس میں فاسفیٹ نمکیات کی موجودگی سے پانی میں کائی کی گروتھ تیز ہوتی ہے؟

(A) کاربونیٹ سالٹس
(B) سلفونک ایسڈ سالٹس
(C) سلفیٹ سالٹس
(D) فاسفیٹ سالٹس

Q10. Ozone is beneficial for us as it absorbs _____.

(A) Infrared radiations
(B) Ultraviolet radiations
(C) Chlorofluorocarbons
(D) Air pollutants

اوزون ہمارے لیے مفید ہے کیونکہ یہ _____ کو جذب کرتی ہے۔

(A) انفرا ریڈ ریڈی ایشنز
(B) الٹرا وائلٹ ریڈی ایشنز
(C) کلورو فلورو کاربنز
(D) ہوا کے پلوٹنٹس

Q11. Which is not a greenhouse effect?

(A) Increasing atmospheric temperature
(B) Increasing food chain
(C) Increasing flood risks
(D) Increasing sea-level

کونسا گرین ہاؤس افیکٹ نہیں ہے؟

(A) فضائی درجہ حرارت میں اضافہ
(B) فوڈ چین میں اضافہ
(C) سیلاب کے خطرات میں اضافہ
(D) سمندری سطح میں اضافہ

Q12. About 99% atmosphere's mass lies within:

(A) 30 kilometer
(B) 35 kilometer
(C) 15 kilometer
(D) 11 kilometer

ایٹموسفیر کا تقریباً 99 فیصد ماس موجود ہے:

(A) 30 کلومیٹر
(B) 35 کلومیٹر
(C) 15 کلومیٹر
(D) 11 کلومیٹر

Chemistry (Subjective) Group - II

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7488

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan]ic equilibrium state reaction does not stop?

(ii) [illegible in source scan] equations: (1+1) (i) $CaCO_3 \rightleftharpoons$ _____ + _____ (ii) $H_2 + I_2 \rightarrow$ _____

(iii) [illegible in source scan]romatic compounds. Give its example. (1+1)

(iv) [illegible in source scan]nctional groups. Give an example of ketonic group. (1+1)

(v) [illegible in source scan]adical is formed?

(vi) [illegible in source scan]omerism and give an example. (1+1)

(vii) [illegible in source scan]uctural formula of glucose and fructose. (1+1)

(viii) [illegible in source scan] between monosaccharides and oligo saccharides. (1+1)

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan] chemical reactions that water is an amphoteric species.

(ii) [illegible in source scan]ses of sulphuric acid.

(iii) [illegible in source scan] of chloroform and carbon tetrachloride.

(iv) [illegible in source scan]yl halides reduced?

(v) [illegible in source scan] Write names of any two ores of copper.

(vi) [illegible in source scan] and matte removed from blast furnace?

(vii) [illegible in source scan]ction for recovery of ammonia in Solvay's process.

(viii) [illegible in source scan]position of petroleum ether? Give its one use.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan]ne layer and where is it found? (1+1)

(ii) [illegible in source scan]ypes of air pollutants are there? Write their names. (1+1)

(iii) [illegible in source scan]o effects of global warming. (1+1)

(iv) [illegible in source scan] sulphur containing compounds emitted naturally?

(v) [illegible in source scan] pesticides and fertilizers cause water pollution? (1+1)

(vi) [illegible in source scan] are responsible for permanent hardness of water?

(vii) [illegible in source scan]hysical properties of water. (1+1)

(viii) [illegible in source scan]ater hampers the cleaning action of soap?

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) [illegible in source scan] vitamins? Name their types. Give importance of the vitamins.

Q5(b). (Marks 05) [illegible in source scan]ilibrium constant expressions for the following chemical reactions: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Q6(a). (Marks 04) [illegible in source scan]s of any five bases.

Q6(b). (Marks 05) [illegible in source scan]rd water? Give disadvantages of hard water.

Q7(a). (Marks 04) What is urea? Describe its raw material and steps for the preparation of urea.

> Note: Q7(b) not visible in source scan.